

The Environmental Biosecurity Project Fund (EBPF) was established in 2018. It provides an annual allocation of \$825,000.

The Environmental Biosecurity Office (EBO) administer the EBPF. We provide leadership, coordination and engagement on national environmental biosecurity issues. We play a key role in strengthening Australia's preparedness and response capabilities.

Scope

EBPF supports projects that improve our ability to prevent exotic environmental pests. We aim to prevent, detect and respond to exotic pests and stop them from becoming established. Exotic pests refer to pests, weeds and diseases that are not yet present in Australia. EBPF is part of a broader system that protects Australia's environment from exotic pests.

A key focus for the EBPF is the Exotic Environmental Pest List (EEPL). The EEPL is a list of 168 exotic species. These species could cause major damage to our environment and way of life. This includes impacts to First Nations heritage sites.

The EBO identifies priority projects that focus on species in the EEPL. This includes improving our understanding and management of the risks of EEPL species. We also consider broader projects to strengthen the environmental biosecurity system.

EBPF objectives

EBPF's objectives are:

- Improve capacity for biosecurity preparedness and response.
- Enhance surveillance, detection and diagnostic capabilities for environmental biosecurity threats.
- Foster expertise, networks and collaboration to enable coordinated and effective responses to environmental biosecurity threats.
- Raise awareness and understanding of environmental biosecurity.

To apply

Download and read the Environmental Biosecurity Project Fund applicant guidelines. Download, complete and submit the Environmental Biosecurity Project Fund application form. These are available below.

We accept proposals throughout the year. However, we encourage you to submit your proposal by 31 October each year. Funding decisions are generally made by the end of each calendar year.

Download

- [Environmental Biosecurity Project Fund Project applicant guidelines \(PDF 254 KB\)](#)

- [Environmental Biosecurity Project Fund Project applicant guidelines \(DOCX 265 KB\)](#)
- [Environmental Biosecurity Project Fund Project application form \(PDF 230 KB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

Completing an EBPF application is an expression of interest process only. Each application will be assessed by our team. The Australian Chief Environmental Biosecurity Officer will make the final decision on whether applications are funded. We will keep you informed as your application goes through this process. We will also progress further actions to comply with Commonwealth requirements to spend money.

Funded projects

EBPF has provided funding for over 50 projects to date. See a list of these projects and some of the important outcomes achieved through EBPF below. Contact the Environmental Biosecurity Office at ACEBO@aff.gov.au for more information.



Case Study: Atlas of Living Australia biosecurity alerts

The Atlas of Living Australia (ALA) is Australia's largest open-source biodiversity data infrastructure. It is often the first platform where new species incursions are publicly recorded in Australia. However, until recently, there was no automated method for notifying biosecurity agencies of these threats.

The ALA developed the Biosecurity Alerts System to notify biosecurity agencies of potential incursions. Since 2020, the ALA has sent more than 1,000 notifications reporting occurrences of possible biosecurity concern to the department.



Case Study: New mobile app to record tree pests

The MyPestGuide® Trees app empowers Australians to report tree pests and diseases. This helps boost early detection efforts and serves as both an identification tool and a reporting platform. The app allows users, including citizen scientists and industry workers, to upload pest images and filter by specific criteria. It assists with pest identification and alerts authorities to exotic pest threats.



Case Study: Enhancing biosecurity through research collaboration

The National Environment and Community Biosecurity Research, Development and Extension Strategy (NECBRDES) 2021-2026 enhances coordination across government, research institutions, industry, and communities. It emphasises setting national priorities, fostering partnerships, and streamlining investments. This work has resulted in more coordinated biodiversity and biosecurity research across all aquatic and terrestrial pests.

Funded projects

Evaluation of the Xylella host status of Australian native plant species

The aim of this project is to identify Australian plants and weeds that could host Xylella. This includes native plants used for amenity, forestry, floriculture, and food crop production.

The data gathered in this project will be used to provide an evidence-based assessment of Xylella host range specifically for Australia to inform Australia's plant biosecurity risk settings, strengthening national preparedness, surveillance targeting and biosecurity risk assessment.

Collaborators: Agriculture Victoria. Investment: \$344,509. Timeframe: May 2026 – June 2028.

In vitro challenge of Australian bat cell lines with the fungal pathogen causing white-nose syndrome

This research will provide critical baseline data for White Nose Syndrome (WNS) risk assessments in Australian bats. Using in vitro bat cell culture models, this project will assess immune responses to the fungus causing WNS. This will inform conservation measures targeted at mitigating the impact of WNS on Australian hibernating bats.

Collaborator: University of Melbourne. Investment: \$128,690. Timeframe: June 2026 – June 2028.

Improving accessibility to specimens of priority plant pests (and related species) within collections

The objectives of this project are to improve access to reference material for species on the Exotic Environmental Pest List (EEPL) and to identify knowledge and specimen gaps, to increase coverage of exotic pests within Australian reference collections, and to and increase national diagnostic capabilities.

Collaborator: CSIRO. Investment: \$70,000. Timeframe: May 2026 – March 2027.

Diagnostic and identification tools for the fireweed (*Senecio inaequidens* - *Senecio madagascariensis*) complex

The project will re-evaluate species boundaries and taxonomic concepts in the globally invasive fireweed (*Senecio inaequidens* – *Senecio madagascariensis*) complex using molecular data and morphology, and based on those findings, develop updated reference data and diagnostic tools and protocols for the complex. This will facilitate efficient and accurate diagnostics and identification of species of the fireweed complex in biosecurity and weed management, including *Senecio inaequidens* which is included on the Exotic Environmental Pest List.

Collaborator: CSIRO. Investment: \$288,688. Timeframe: May 2026 – January 2028.

A National Diagnostic Protocol for *Teratosphaeria* canker, stem canker and leaf blight (*Teratosphaeria* spp.)

This project aims to deliver validated national diagnostic protocols (NDP) for accurate identification of the foliar pathogen, *Teratosphaeria destructans*, and the stem canker pathogens, *T. gauchensis* and *T. zuluensis*.

The project aims to deliver a protocol that can be used to identify and distinguish exotic eucalyptus leaf blight and canker species from already established/endemic taxa. The project also aims to develop and validate species-specific molecular diagnostic tools that can be implemented in the resulting NDPs, as *Teratosphaeria* species are notoriously slow growing and can challenge conventional diagnostic methods, particularly in terms of time to complete a diagnosis.

Collaborator: NSW Department of Primary Industries and Regional Development. Investment: \$362,581. Timeframe: April 2026 – May 2029.

Priority Actions of the Invasive Ant Biosecurity Plan

Project Overview

Australia's National Invasive Ant Biosecurity Plan 2018–2028 identifies several priority actions required to strengthen the prevention of exotic ant incursions and establishment, as well as to improve eradication outcomes. This project will deliver three of these priority actions. First, it will generate a dichotomous key for all species within the *Monomorium salomonis* species group. Second, it will develop a national protocol document outlining best-practice treatment and post-treatment assessment methods for invasive ants targeted by eradication programs. Third, it will undertake a taxonomic revision of the genus *Ochetellus*, with a particular focus on species occurring outside Australia.

Collaborator: Myrmex Pty Ltd. Value: \$212,522.00. Timeframe: Feb 2026 – Jan 2028. Status: Active

The Bug Hunt Phase 2: implementation of citizen science-based biosecurity reporting through community education

Project Overview

This project will test and build the knowledge, skills, and networks needed to establish a long-term citizen science program that strengthens Australia's environmental biosecurity surveillance. It will put in place the recommendations from the first phase of Bug Hunt 1: Developing Communities of Practice for General Surveillance – Environmental Biosecurity Reporting. Citizen science already plays a key role in biosecurity (2022–24 Environmental Biosecurity Project Fund – DAFF). This project will introduce new strategies to engage citizens in monitoring invasive invertebrates.

Building on the strong foundation of the initial project, this next phase will explore three key approaches to better understand how to expand and strengthen citizen science surveillance efforts:

- School Education Program – [Bug Hunt Education Resources | Cool.org](#), including in-person promotion, development of curriculum-appropriate lesson plans and worksheets, and delivery of training to teachers.
- Geographic Targeting – Focused community participation based on biosecurity risk, particularly near ports of entry.
- National “Bugs in My Backyard” (BIMBY) Week Trial – [Bug Hunt – Become a bug detective today](#), aimed at growing participation across Australia, similar to the [Great Aussie Bird Count](#).

Outcome Indicators

- Level of adoption of Bug Hunt and environmental biosecurity messaging by partner citizen science participants and organisations.

- Number of reports and sign-ups using approaches identified in Stage 1 of the Biosecurity Bug Hunt project.
- Updated and expanded understanding of the challenges and benefits of different diagnostic apps, including investigation of challenges identified in the previous Bug Hunt project and proposed solutions.
- Improved capability of AI diagnostics to better identify invasive species.

By achieving these outcomes, the project will contribute to protecting Australia's wildlife, industries, and communities.

Collaborators: Invasive Species Council Inc., Invertebrates Australia. Investment: \$401,610.

Timeframe: September 2025 – June 2027.

Development of a targeted surveillance data platform for the National Avian Influenza Wild Bird (NAIWB) Surveillance Program

Funding from the Office of the Chief Environmental Biosecurity Officer is supporting Wildlife Health Australia and Epi-Interactive to build a specialised electronic data platform for targeted surveillance data from the NAIWB Surveillance Program (<https://wildlifehealthaustralia.com.au/Our-Work/Surveillance/Wild-Bird-Surveillance>) This program tracks strains of avian influenza viruses found in apparently healthy wild birds around Australia. The platform will improve data management and sharing of information on avian influenza viruses through:

- streamlining the data upload process – allowing data from surveillance partners to be uploaded efficiently and consistently,
- developing electronic data validation – ensuring data integrity and efficient data management by verifying data accuracy and quality against pre-defined rules,
- providing a single, centralised location for data storage – data from surveillance partners will be stored in a single secure location, improving efficiency for data analysis and reporting,
- maintaining secure data access – only those with appropriate authorisation will be able to access the data.

In achieving all of this, this project will help protect Australia's wildlife, industries, and community.

Collaborators: Wildlife Health Australia. Investment: \$150,000. Timeframe: October 2024 – July 2026.

Cultural wisdom protecting forests and Country

This project will establish on-ground networks to increase our national capacity to detect and report the presence of exotic pests in our natural environments. It will help protect our unique biodiversity from forest pest and disease threats such as myrtle rust.

The five-year project will identify opportunities to create and introduce tools for Indigenous communities to manage invasive pests and diseases on Country. It will build capacity and establish regional coordinators to engage Indigenous rangers. This will foster relationships and establish formal networks between ranger groups and develop technical expertise.

The project will deliver on-ground activities, such as reporting, surveillance and monitoring programs. A steering committee will help ranger groups prioritise their objectives and provide feedback on project deliverables.

Collaborators: Department of Climate Change, Energy, the Environment and Water (NSW), Department of Agriculture and Fisheries (Qld) Investment: \$988,816 Timeframe: Aug 2024 - Mar 2029.

ALA – cofounded software and database developer: Expanding ALA biosecurity capabilities

This project will support the continued delivery and enhancement of the National Biosecurity Data system and thereby enable easier access to relevant and timely data to support biosecurity operations and research. The project will continue the existing collaboration between the Atlas of Living Australia (ALA) and the Department of Agriculture Fisheries and Forestry's (DAFF) Environmental Biosecurity Office (EBO) through the contribution of salaries to support the ALA's Biosecurity Alerts Service.

Collaborators: CSIRO. Investment: \$80,000. Timeframe: Jun 2024 - Apr 2025.

PaDIL drafting and upload of native ant images and biological descriptions

The aim of this project is to develop diagnostic descriptions for ant species found in Western Australia and Queensland. The diagnostic images along with the diagnostic descriptions will be made available for use on the Pest and Disease Image Library (PaDIL) a publicly available database of diagnostic quality images supporting Australia's biosecurity system.

Collaborators: Department Of Primary Industries and Regional Development (WA). Investment: \$28,403. Timeframe: May 2023 - Jul 2024.

National Environmental and Community Biosecurity RD&E Strategy (NECBRDES) - National Coordinator: Phase 2

The duties of the National Coordinator will be centred upon the delivery of objectives laid out in the NECBDRES and will build upon the work undertaken in the previous 12 months as described in the NECBRDES Transition Plan as Phase 1 of NECBRDES implementation.

The main activities will be the formation of biosecurity cross-sectoral working groups to understand and identify RD&E requirements and objectives across the biosecurity sectors of animal, plant, environmental and marine biosecurity. The National Coordinator will manage the working groups to deliver mutually beneficial outcomes for the biosecurity sectors and in particular to progress environmental and community biosecurity RD&E work.

Collaborators: Wild Matters Pty Ltd. Investment: \$350,000. Timeframe: Jun 2023 - May 2025.

National Avian Influenza Wild Bird Surveillance Program 2023-2026

This project aims to undertake additional sampling of wild birds under the National Avian Influenza Wild Bird Surveillance Program (NAIWBS) 2024-26. The project will support surveillance for influenza viruses in wild birds to maintain diagnostic test capability, laboratory capacity and field sampling capability.

This program supports Australia meeting national and international commitments relating to trade and reporting of avian influenza to the World Organisation for Animal Health.

Collaborators: Wildlife Health Australia. Investment: \$150,000. Timeframe: Jan 2024 - Dec 2026.

Development of Atlas of Living Australia as a central biosecurity database

The Atlas of Living Australia (ALA) aggregates species identity and location data through various methods including direct API linkages, bulk transfer, and from citizen science applications with mobile capability such as Bio-Collect and iNaturalist. The ALA is a unique capability and database repository that is continually being updated with species identifications and locations from throughout Australia.

A new capability has been developed that will allow the ALA to automatically inform the Department of Agriculture, Water and the Environment of any new information concerning the majority of species on the National Priority List of Exotic Environmental Pests, Weeds and Diseases that are discovered in Australia through one of ALA's network channels.

Collaborators: CSIRO. Investment: \$80,000. Timeframe: Apr 2023- Mar 2024.

National Environmental and Community Biosecurity RD&E Strategy (NECBRDES) – National Coordinator

The engagement of a National Coordinator to progress the objectives outlined in the National Environment and Community Research, Development and Extension Strategy 2021-2026 (NECBRDES).

The main activities will be the formation of biosecurity cross-sector working groups to understand and identify RD&E requirements and objectives across the biosecurity sectors of animal, plant, environmental and marine biosecurity.

Collaborators: Wild Matters Pty Ltd. Investment: \$100,000. Timeframe: Jun 2022 - May 2023.

Building contemporary molecular diagnostic capability for northern Australia

Molecular diagnostic capabilities for the detection and identification of environmental biosecurity pests, weeds and diseases form a central part of strategic plans for exotic environmental biosecurity. Unregulated pathways involving migratory animals and airborne pests, in addition to huge areas to

manage the risk of pest and disease incursions. The capacity to detect the presence of a pest without actually seeing it using molecular diagnosis will help address these northern Australia challenges.

This project will build contemporary molecular diagnostic capability and develop DNA reference collections in northern Australia. This will be used immediately to support environmental and plant biosecurity risk management. The project will also improve familiarity and promote use of this technology through training activities and workshops. The outputs will be integrated with other national initiatives that seek to standardise biosecurity molecular diagnostic methods and systems.

Collaborators: Northern Territory Department of Industry, Tourism and Trade, Department of Agriculture, Fisheries and Forestry – Northern Australian Quarantine Strategy. Investment: \$200,000. Timeframe: May 2021 - May 2024.

Society and Science: A new approach to wildlife disease surveillance

Early detection of new and emerging wildlife diseases with an effective surveillance system is important for preventing native animal population declines and wildlife-species extinctions.

This project aims to develop an effective NSW wildlife disease surveillance system, as a national model, with broad spatiotemporal and taxonomic coverage. This will act as a direct real-time conduit between public observations of significant wildlife disease and responding agencies. The system will be designed through robust sociological research to ensure the needs of diverse expert and community participating stakeholders are taken into account and to maximise its user appeal.

The composition of the project and the endpoint objectives will also reveal how participation in wildlife disease surveillance influences the perceptions and values of wildlife health in our community.

Collaborators: Charles Sturt University. Investment: \$231,772. Timeframe: Jun 2021 - Aug 2026.

An operational molecular pipeline to detect reptile species common in the Illegal wildlife trade

This project addresses the lack of reptile specific detection methods for tackling the illegal live pet trade, specifically, focussing on cases where the specimens have already been moved, hidden, or destroyed. The project will deliver two sequential standard operating procedures (SOPs) that can detect trace reptile signatures in a biosecurity setting. The first SOP will enable a novel application of Loop Mediated Isothermal Amplification (LAMP) based colorimetric detection methods. The second SOP will permit laboratory-based validation, reminiscent of the pipeline employed by onsite presumptive drug detection equipment.

Development of this pipeline, and subsequent translation into operational practice, is a key biosecurity priority to detect and identify reptile species and thereby curb environmental and agricultural threats posed by smuggled invasive animals.

Collaborators: The University of Adelaide. Investment: \$35,100. Timeframe: Feb 2022 - Feb 2024.

Creating and maintaining annotated checklists of introduced and invasive species across Australia

This project will develop validated and verified annotated checklists of introduced and invasive species for a suite of geographical entities within Australia. These checklists will form the basis for an Essential Biodiversity Variables (EBV) triage tool and baseline data on the introduction, occurrence, and impacts of alien and invasive species that can be disaggregated nationally, territorially and by protected areas and islands.

The checklists will be compatible with the Atlas of Living Australia (ALA), an important aggregator of biosecurity and biodiversity datasets. The Global Register of Introduced and Invasive Species (GRIIS) lists for Australia will be updated and incorporated into the ALA and used as a quantitative tool for refined monitoring and reporting of invasive species impacts.

Collaborators: Biodiversity Data Management Ltd., CSIRO - Atlas of Living Australia, La Trobe University, Australian Bureau of Agricultural and Resource Economics and Sciences. Investment: \$134,860. Timeframe: Feb 2022 - May 2024.

2022-2032 National Roadmap for Wildlife Health Research, Development and Extension

A ten-year roadmap for wildlife health research, development, and extension (RD&E) will guide and inform an effective and nationally coordinated RD&E approach that secures sustainable resourcing. This strategy will facilitate the finding of effective solutions to address major wildlife health issues. Wildlife health problems within Australia, and abroad, are a serious threat to our natural environment, health, economy, and way of life.

The drafting of the roadmap will begin with a review of published literature and conversations with stakeholders and research providers to identify their needs. This will inform national research and development priorities, capacity building requirements and preferred methods for information sharing. The roadmap will incorporate findings from recent disease prioritisation projects such as the [National Priority List of Exotic Environmental Pests, Weeds and Diseases](#) to further inform research, development, and extension priorities. A research priority may involve targeted innovation of current surveillance activities to ensure their detection and adequate response to a new wildlife disease threat.

Collaborators: University of Melbourne, University of Tasmania, Wildlife Health Australia, Charles Sturt University, University of Sydney, Deakin University, Macquarie University. Investment: \$195,000. Timeframe: May 2021 - Jun 2022.

Jointly funded position to support biosecurity data flow and interoperability

This project will co-fund a CSIRO staff member who will work between the Atlas of Living Australia (ALA) and the Environmental Biosecurity Office. The role will deepen the existing partnership and

build new collaborations in the biosecurity information sector in a coordinated manner while it supports the joint program of work between the two organisations.

Collaborators: Department of Agriculture, Fisheries and Forestry, CSIRO – Atlas of Living Australia.
Investment: \$80,000. Timeframe: May 2022 - Jun 2023.

Developing Communities of Practice for General Surveillance - Environmental Biosecurity Reporting

Citizen science already plays an important role in biosecurity detections. With the right incentives and supporting systems there is potential for community assisted detections of insects and other visible species of national concern. The target species of concern will be drawn from the National Priority List of Exotic Environmental Pests, Weeds and Diseases and species identified by the joint Invasive Species Council and Monash University Invasive Insects Risks and Pathways Project.

The project aims to use invasive insects of biosecurity concern to trial how environmental biosecurity reporting and awareness for newly established and emerging pest species within Australia can be improved, increased, and streamlined through the use of citizen science. Citizen science is an ever-growing field of public participation in data collection. This project will demonstrate whether this large resource can be leveraged to build capacity for environmental biosecurity surveillance and reporting.

Collaborators: Invasive Species Council Inc., La Trobe University, CSIRO, Australian Citizen Science Association. Investment: \$232,100. Timeframe: May 2022 - May 2024.

Indigenous Forest Health and Environmental Biosecurity

This project collaboratively developed and delivered an environmental biosecurity training program with the purpose of improving the capacity of Indigenous Rangers, Natural Resource Management (NRM) and other land managers across Australia. The training program will increase awareness and substantially expand our national capacity to detect and report on exotic environmental pests, including plant pathogens that threaten Australia's forests and culturally significant species.

The project also delivered environmental biosecurity and forest health workshops, including specific myrtle rust training to Indigenous Rangers and landowners in NSW and QLD to increase engagement and improve capacity to detect and report on biosecurity threats. It provided tools and knowledge to help protect traditional lands from exotic and established pest and disease threats. A short film was developed to further promote management capabilities, drawing upon Indigenous cultural values of forest and plant species that are threatened by exotic pests and pathogens such as myrtle rust.

Collaborators: Department of Planning and Environment, NSW, Department of Agriculture and Fisheries, QLD, Department of Primary Industries, NSW, Plant Health Australia (PHA), Queensland Parks and Wildlife Services, Department of Environment and Science, Department of Agriculture, Fisheries and Forestry and Natural Resource Management (NRM) Knowledge Broker. Investment: \$404,493. Timeframe: May 2022 - Mar 2024.

Effects of *Ehrlichia canis* on wild dingoes

Ehrlichiosis is a bacterial disease that affects dogs in tropical and sub-tropical areas around the world and is transmitted by the tick *Rhipicephalus sanguineus* (brown dog tick). Multiple tick species have been found on dingoes. Therefore, there is concern that *Ehrlichia canis* may enter wild dingo populations and cause detrimental impacts on dingoes.

The overall goal of this project is to determine the prevalence of brown dog ticks and *E. canis* in wild dingoes inhabiting areas nearby to where *E. canis* has been detected in domestic dogs and confirm any health-related impacts to infected animals. The approach that will be used to achieve the aims of this project is to coordinate the collection of blood, ticks, and tissue samples from dingoes during lethal control programs near target areas in Northern Australia. Blood samples and ticks will be sent for assessment to collaborators in Townsville and tissue samples will be sent for assessment to collaborators in Adelaide.

The project did not proceed as there were logistical issues affecting access to sufficient dingo samples and was assessed that the project would not be able to deliver the expected outputs within a reasonable timeframe.

Collaborators: University of Southern Queensland, James Cook University, ZoolGenetics Pty Ltd., Aussie Feral Control, ABS Scrofa Pty Ltd. Investment: \$188,690. Timeframe: Jun 2022 - Dec 2023.

ALA ongoing operation and maintenance of automated alerts

The Atlas of Living Australia is a unique capability and database repository that is continually being updated with species identifications and locations throughout Australia. A new capability has been developed that will allow the ALA to automatically inform the Department of new information concerning many species on the National Priority List of Exotic Environmental Pests, Weeds and Diseases that are discovered through one of the ALA's network channels. The project delivers system maintenance and access to the Atlas of Living Australia pest report notification system, updates to notifiable species lists and training and service support for system functionality.

Collaborators: CSIRO Atlas of Living Australia. Investment: \$22,726. Timeframe: Jun 2021 - Aug 2022.

Assessing the Invasiveness Risk of Non-Indigenous Fish in the Australian Ornamental Trade – Implementing the Risk Assessment Tool

The project will assess the biosecurity and invasiveness risk of over 500 species of non-native ornamental fish including identification of overall biosecurity risk level, likelihood of release, likelihood of invasion and assessment of consequence. A risk assessment tool developed by the South Australian Research Development Institute will be used to evaluate species and justify the implementation of sound policy, regulation and operational decisions relating to the potential biosecurity risk of exotic ornamental fish across all Australian jurisdictions. Specifically, it will provide robust support for the potential for increased regulation of species deemed to be of high risk in the Australian ornamental aquarium trade by reducing their potential to become further established and a future biosecurity issue.

Collaborators: South Australian Research Development Institute (SARDI), the Environment and Invasives Committee (EIC), Freshwater Vertebrate and Invertebrate Working Group (FVIWG).
Investment: \$123,000. Timeframe: Jan 2021 - Jun 2022.

Non-Indigenous Freshwater Fish and Invertebrate Plan and national communications plan

The project will deliver a National Non-Indigenous Freshwater Fish and Invertebrate Plan (the plan) and national communications plan. The plan will propose management actions to mitigate the risk of environmental impacts to Australia's natural waterways by providing a consistent and cohesive approach to identify, manage and regulate non-indigenous freshwater fish (including aquarium pets) and invertebrates in Australia. The communications plan will develop a national approach and package of material that can be used by the national communications network and distributed through industry channels.

Collaborators: Queensland Department of Agriculture and Fisheries (QDAF), Freshwater Vertebrate and Invertebrate Working Group (FVIWG). Investment: \$65,000. Timeframe: Mar 2021 - Apr 2022.

Spotted lanternfly biology, ecology and awareness in the Australian environment

Spotted Lantern fly (*Lycorma delicatula*) is an emerging pest of agricultural, urban and environmental concern. The pest attacks over 70 plant species, primarily woody hosts. Cesar Australia will undertake a desktop study to address preparedness and awareness for Spotted Lantern fly (SLF) as a pest of biosecurity concern in the Australian environment. Deliverables include a review of SLF biology, invasive pathways and management, and a host plant list and distribution map within Australia. This project provides the opportunity to find and improve the information available on environmental and social amenity impact of this species. The project is co-funded by the Australian Plant Biosecurity Science Foundation.

Collaborators: Cesar Australia, Australian Plant Biosecurity Science Foundation. Investment: \$36,365. Timeframe: Oct 2020 - Apr 2021.

Browsing ant workshop and support for the NBMCC

In March 2020 the Department of Agriculture, Water and the Environment engaged Natural Decisions Pty Ltd to analyse, assess and report on multiple national eradication response programs for Browsing ants (*Lepisiota frauenfeldi*) against the National Environmental Biosecurity Response Agreement (NEBRA) criteria. Natural Decisions subsequently developed three tools, 1) a national significance assessment tool, 2) a benefit costs analysis (BCA) tool and 3) a surveillance tool.

This project will deliver a workshop on the use of tools, key concepts, and their practical application for National Biosecurity Management Consultative Committee (NBMCC) members and Secretariat, the NEBRA Custodian, and nominated other technical experts. The project will also provide support

material and help desk support following the workshop for general advice and assistance with the use of the tools as well as the review of draft assessments for other invasive ant species.

Collaborators: Natural Decisions Pty Ltd. Investment: \$19,580. Timeframe: Feb 2021 - Mar 2022.

Biosecurity Commons – Virtual Biosecurity Lab

The project will operationalise the world's first 'Biosecurity Virtual Lab' for use in decision making. It is anticipated that the project will enable researchers to share modelling and analytics in a secure space, improving transparency and the ability to trace model and data provenance, while building trust and improving confidence in model outputs. Researchers will be able to use this platform to produce models such as risk mapping, surveillance design, species dispersal, impact estimates and resource allocation, proof of freedom using a series of established workflows. Training and engagement will be undertaken – to enable the decision-making tool to be taken up and used in agencies. It is important to note, this project is a pilot, and all funding is only available until the end of 2022-23. Sustainable funding options for the ongoing operation of the decision-making tool will be considered as the project progresses

Collaborators: University of Melbourne, Griffith University, Queensland Department of Agriculture and Fisheries, Australian Data Research Commons (ARDC). Investment: \$187,500. Timeframe: May 2021 - May 2023.

Vulnerability of Australian bats to White Nose Syndrome

This project will determine the vulnerability of Australian bats to white nose syndrome (WNS). Completion of this project is expected to generate urgently needed, species-specific and spatially explicit wildlife health management recommendations, and to provide the first detailed information about the winter ecology and hibernation biology of Australian cave-roosting bats. WNS is ranked in the top five of native animal diseases and their pathogens and listed in the national priority list for exotic environmental pests and diseases. The likelihood of WNS being translocated to Australia is 'almost certain' over the next ten years.

Collaborators: Western Sydney University Australian Research Council Linkage Project. Participating organisations: The University of New England, University of Winnipeg, Taronga Conservation Society Australia, The State of Victoria through its Department of Environment, Land, Water and Planning (DELWP), Zoos Victoria, Wildlife Health Australia Incorporated, Australasian Bat Society Inc, Australian Speleological Federation Incorporated, NSW Department of Planning, Industry and Environment – National Parks and Wildlife Service, University of Melbourne. Investment: \$81,818. Timeframe: Apr 2020 - May 2023.

Low Energy X-ray Seed Automatic-Detection Algorithms (Phase Two)

After successfully completing Phase One of the Proof of Concept (POC) project to evaluate the feasibility of using low-energy 3D X-ray for high resolution imaging and automated detection of seeds in incoming packages entering Australia. Phase Two will carry out the system installation at a mail centre or training facility, system integration at the mail centre, seed detection algorithm and seed packet biometric

optimisation followed by full system performance assessment for automatic identification of seed packets in the mail pathway. The project will help to manage environmental biosecurity risks that could enter Australia through illegally imported seeds. The image data already collected by the system will also be inspected for other items such as currency and narcotics.

Collaborators: Rapiscan Systems Pty Ltd and Taronga Zoo, Department of Agriculture, Water and the Environment Biosecurity Innovation Program - Modern Seamless Border Clearance Project.

Investment: \$676,756. Timeframe: Jan 2021 - Jul 2021.

National Myrtle Rust Action Plan implementation

Myrtle Rust is capable of infecting 358 native plant species in Australia, posing a serious threat to Australia's native biodiversity and ecosystems. This project will design and develop a pathway to implement the 'Myrtle Rust in Australia A draft Action Plan'. The Plant Biosecurity Science Foundation will bring together a panel of experts comprising Australia's three foremost Myrtle Rust experts, an ecologist, rust disease specialists, a vegetation dynamics specialist, and an indigenous representative to assess the draft action plan. The project will deliver a report endorsed by the Panel that: provides an assessment of the Action Plan, highlighting the key priorities for action and investment by governments; estimates the potential costs of myrtle rust in the absence of co-ordinated national action and makes a recommendation on process, time and resources required to implement the Action Plan to minimise the negative impacts of myrtle rust in Australia.

Collaborators: Plant Biosecurity Science Foundation. Investment: \$28,100. Timeframe: Feb 2021 - Jun 2021.

Risk Assessments for Established Vertebrate Pests

Funding for a suitably qualified person to coordinate risk assessments for the classification of threat categories of non-indigenous vertebrate animal species for addition to the list published and maintained by the Environment Invasives Committee the Australian List of Threat Categories of Non-indigenous Vertebrates.

Collaborators: Department of Primary Industries and Regions South Australia. Investment: \$25,000. Timeframe: Apr 2021 - Apr 2022.

Australian Vertebrate Pest Conference 2021 sponsorship

The Australasian Vertebrate Pest Conference (AVPC) has a long and prestigious history as one of the regions keystone biosecurity related conferences bringing together practitioners, researchers, wildlife managers and policy advisors to network, share their stories and discuss the future of vertebrate pest management in Australia and beyond. This event was originally scheduled to be held in May 2020, but due to COVID-19, it has been postponed to late 2021.

Collaborators: Centre for Invasive Species Solutions, Established Pest Animals and Weeds Management Pipeline Program (DAWE). Investment: \$20,000. Timeframe: May 2020 - May 2020.

Australasian Weeds Conference 2021 sponsorship

The Australian Weeds Conference will be running for its 22nd year and is a significant bi-annual conference for biosecurity, weed management, research and policy stakeholder from Australia, New Zealand and other countries in our region. It will provide insights into cutting-edge weed science and research, local and national innovative policy, new operational practices and tools, best practice on-ground management, chemical innovations, alternative weed control and on-ground case studies that help to protect Australia's agricultural and environmental landscapes from the threat of weeds. This event was originally scheduled to be held in October 2020, but due to COVID-19, it has been postponed to late 2021.

Collaborators: The Council of the Australasian Weed Societies (CAWS), together with local host the Weed Management Society of South Australia (WMSSA). Investment: \$23,000. Timeframe: Oct 2021 - Oct 2022.

Australian Wildlife Health Institute workshop

The University of Melbourne Australian hosted and run a workshop to discuss the establishment of a 'Wildlife Health Institute (AWHI)'. The intent of AWHI is to integrate and formalise wildlife health arrangements across disciplines to devise practical and effective solutions that balance the health of wildlife and ecosystems on which we depend with human, societal, agricultural and economic needs.

It is proposed that AWHI will establish and lead a collaborative research and training alliance that will leverage and build on existing research and training capacity across Australian universities and research institutes to carry out research and training that is structured in a translational framework for the deployment of effective solutions for prioritised wildlife health problems. AWHI will create unified resources responsive to the needs of both policy makers and practitioners and will emphasise communication and engagement with both professional and public audiences.

The need for a strategic research and training approach for wildlife health and a dedicated structure to support it were the major recommendations from the Wildlife Health Australia Universities Workshop Report in 2007. This workshop acts on those recommendations and addresses the increasing need demonstrated since then.

Collaborator: The University of Melbourne. Investment: \$7,017. Timeframe: Nov 2019 - Nov 2019.

Illegal Wildlife X-ray Algorithm

To keep pace with increases in volumes of trade, especially in the international mail and express freight pathways favoured by wildlife smugglers, the department is looking to invest in smarter ways of screening for wildlife/animals. This project will develop automated 3D algorithms for the detection of illegally traded animals and wildlife in passenger and mail pathways. By partnering with Taronga Zoo, the project will bring together expertise in animal conservation, veterinary medicine, imaging and computer science to deliver a program that has a high probability of success in delivering a significant step towards reducing wildlife trafficking through the borders in Australia and New Zealand.

Collaborators: Rapiscan Systems Pty Ltd and Taronga Zoo, Department of Agriculture, Water and the Environment Biosecurity Innovation Program - Modern Seamless Border Clearance Project.
Investment: \$238,000. Timeframe: Jun 2020 - Jun 2021.

Western Australia Ant Imagery Project

This project will generate high quality images of local and endemic ant species in Western Australia to assist in identification accurate and quick recognition of exotic ant incursions.

The high quality verified diagnostic materials that will improve biosecurity preparedness and response by supporting faster elimination of common local species of ants from trap and other samples. The images will facilitate faster eradication response times and provide materials for use in communicating ant-related issues to the general public

Collaborator: Western Australian Agriculture Authority. Investment: \$36,000. Timeframe: Jun 2020 - Jul 2020.

Atlas of Living Australia Biosecurity Alert Notification

The Atlas of Living Australia (ALA), is a National Collaborative Research Infrastructure Strategy (NCRIS) database hosted and maintained by the CSIRO. It groups observational data (such as species identifications from throughout Australia) from many sources including museums and collections, government organisations, research organisations, and the citizen science sector.

This collaborative project between the Department and CSIRO will pilot digital protocols and business processes for information sharing and will enable the ALA to automatically inform the Department of any potential species of interest that are discovered in Australia through one of ALA's network channels. The project will enhance our biosecurity surveillance coverage and further improve early intervention of exotic-species incursions.

Collaborator: Atlas of Living Australia CSIRO. Investment: \$35,000. Timeframe: Jun 2020 - Jun 2021.

Environmental Biosecurity Risk Mitigation Plan of Australian Native Bees

Australia is home to over 1,500 species of native bees. Alongside our hardworking European honey bees, our native bees play an important role in pollinating both native plants and commercial crops such as mango, blueberry, eggplant, tomato, almonds and macadamia. Pests and diseases of bees not only have the potential to devastate bee colonies, but may also impact on the health of native plants and ecosystems should our bees be unable to pollinate them. This project is investigating the potential biosecurity risks and pathways that may allow exotic bees or pests of Australian native bees to enter Australia. A plan to mitigate against these identified risks that is fit for purpose and links in with existing programs will be developed to guide what measures may be required to protect native bee populations as best as possible should an incursion occur.

Collaborators: Plant Health Australia, NSW Department of Primary Industries, Dr John Roberts, Honey bee pathogen research scientist, CSIRO, Associate Professor Robert Spooner-Hart, Western

Sydney University, Dr Dean Brookes, University of Queensland, Professor James Cook, Hort. Innovation Frontiers Pollination Program, Western Sydney University. Investment: \$156,142. Timeframe: Mar 2020 - May 2022.

Marine Pest Identification and Engagement Photo Album

Managing surveillance of Australia's marine environment is a major challenge. The environment spans over 59 000 kilometres and attracts a diverse range of recreational and commercial users. Each user has the potential to contribute significantly to our national surveillance capability.

This project brings together a marine pest photo album of priority marine pest species to develop education with awareness materials for marine pest observer groups and other stakeholders. The album will also be used as a source of photos for other departmental activities including awareness and education.

The project also supports Australia's national strategic Marine Pest Plan 2018-2023 for marine pest biosecurity and improve both passive surveillance and awareness of marine pest biosecurity risks, management actions and shared responsibilities.

Collaborator: Biofouling Solutions Pty Ltd. Investment: \$55,780. Timeframe: Mar 2020 - Jul 2020.

Mobile applications for identifying and reporting pest of forests and trees

Australia's forests encompass native, planted and urban forests and represent the world's seventh largest forest estate - managed and used by over 100 stakeholder groups. This project will enable the development of the MyPestGuideTREES App – a mobile surveillance application that combines forest/tree pest information with a reporting capability. This capability was identified as a gap in the National Forest Biosecurity Surveillance Strategy 2018-2023 and associated

Implementation Plan developed jointly by Department of Agriculture, Water and the Environment and the Australian Forest Products Association (AFPA).

This project will bring together expertise from state and territory governments, Plant Health Australia and forest sector partners to strengthen national surveillance capability and capacity amongst forest, plantation and urban tree user and manager groups.

Collaborators: Plant Health Australia; Forest and Wood Products Australia; Department of Agriculture, Water and Environment; WA Department of Primary Industries and Regional Development; Primary Industries and Regions SA; Agriculture Victoria; TAS Department of Primary Industries, Parks, Water and the Environment; NSW Department of Primary Industries; QLD Department of Agriculture and Fisheries; and, NT Department of Primary Industries and Resources. Investment: \$70,000. Timeframe: May 2020 - Oct 2022.

National forest pest surveillance training package

Plant Health Australia is working with a range of collaborative partners, including Forest and Wood Products Australia and state government agencies to produce a nationally consistent forest/pest biosecurity training program accessible to key stakeholder groups via PHA's Biosecurity Online Training (BOLT) platform. The training package will support effective use of the MyPestGuideTREES app and develop a set of core surveillance training materials that can be contextualised for specific stakeholder groups.

This project will support objectives identified in the National Forest Biosecurity Surveillance Strategy 2018-2023 and associated Implementation Plan developed jointly by Department of Agriculture, Water and the Environment and the Australian Forest Products Association (AFPA).

Collaborators: Plant Health Australia; Forest and Wood Products Australia; Department of Agriculture, Water and Environment; WA Department of Primary Industries and Regional Development; Agriculture Victoria; TAS Department of Primary Industries, Parks, Water and the Environment; NSW Department of Primary Industries; QLD Department of Agriculture and Fisheries; and, NT Department of Primary Industries and Resources. Investment: \$100,000. Timeframe: May 2020 – Feb 2023.

Assessment of Browsing ant (*Lepisiota frauenfeldi*) under the National Environmental Biosecurity Arrangement (NEBRA)

This project will deliver expert analysis, assessment and reporting on multiple national eradication response programs for Browsing ants (*Lepisiota frauenfeldi*) against the NEBRA criteria of national significance, level of surveillance activity required to provide confidence of eradication, and Benefit cost analysis. The project will also examine national environmental impacts and cost implications posed by Browsing ant and deliver assessment methodologies/guidance materials that can be used to guide other NEBRA programs in future.

Collaborator: Natural Decisions. Investment: \$89,056. Timeframe: Feb 2021- Mar 2022.

Development of a National Diagnostic Protocol (NDP) for *Ceratocystis fimbriata* and related taxa

The National Priority List of Exotic Environmental Pests, Weeds and Diseases (EEPL) currently lists three species of *Ceratocystis* as a risk to Australia's environment and amenity.

This project will develop Australia's diagnostic capability to detect the fungal plant-pathogens *Ceratocystis* spp using genomic based methods. The development of this diagnostic capability is highly specialised and requires botanical, mycological and molecular biological expertise. The end product will be made available nationally via the National Diagnostic Protocol (NDP) database.

The development of an NDP for the identification of *Ceratocystis* spp will assist in the identification and detection of the pathogen which can affect the health of both native species (*Acacia* and *Myrtaceae*) and production crops such as forestry, mangoes and sweet potatoes.

Collaborators: Plant Health Australia, Department of Agriculture and Fisheries, Queensland (DAF Qld), the University of Queensland (UQ) and the Forestry Agricultural Biotechnology Institute at the University of Pretoria, South Africa (FABI). Investment: \$22,727. Timeframe: May 2020 - Apr 2022.

Supporting Australia's leadership in the Intergovernmental Platform on Biodiversity and Ecosystem Services (IPBES) thematic assessment on invasive species and their control

This project supports Australia's representation on the Intergovernmental Platform on Biodiversity and Ecosystem Services (IPBES) Thematic assessment on invasive species and their control between 2019 and 2022. The completed report will generate policy options for invasive pest species management in Australia and globally, and provide an independent, expert perspective on the impact invasive pests may have on the Australian environment and agricultural sector. These insights will further the development of strategies and procedures for forecasting, preventing and controlling the spread of invasive alien species.

Collaborator: CSIRO. Investment: \$29,484. Timeframe: Aug 2019 - Apr 2022.

Northern Australian freshwater aquarium trade scoping study

The aquarium trade poses a major biosecurity risk to endemic freshwater ecosystems internationally and in Australia. This project addresses these identified issues through conducting a scoping study for improving understanding and management of aquarium trade related biosecurity risks in Northern Australia. To date this project has engaged key stakeholders to provide perspective and expertise into the development of the strategy. A draft report is currently being reviewed. This study will be used by the Northern Australian Quarantine Strategy (NAQS) to provide guidance around the development of surveillance activities and build response and preparedness capabilities.

Collaborator: James Cook University. Investment: \$15,000. Timeframe: Mar 2020 - Apr 2020.

National Environmental Biosecurity Response Agreement (NEBRA) Incidents Portal

Effective responses to environmental biosecurity incidences using NEBRA require readily access to key information. This project develops a web-based portal that will be a central repository of information for members of the consultative committees. These committees form in response to specific incidents and decide if it is technically feasible to eradicate a pest or disease. They are the National Biosecurity Management Consultative Committee (NBMCC) and Consultative Committee on Introduced Marine Pest Emergencies (CCIMPE).

Collaborator: Plant Health Australia. Investment: \$146,295. Timeframe: Apr 2019 - Mar 2020.

Environmental Biosecurity Plan for Australian Acacia Species

Acacia is the largest genus of flowering plants in Australia and is vulnerable to exotic pests that could have significant environmental impact on the keystone species. This project will finalise the Environmental Biosecurity Plan for Australian Acacia Species and produce a Biosecurity Implementation Plan, including priorities, actions and possible responsible organisations/ groups to carry out activities. It will also develop the high priority pest list for Acacia species, deliver a map of key stakeholders and identify risk pathways. The plan will describe various activities that can be undertaken to improve Australia's ability to respond to the introduction of new Acacia pests.

Activities to date include stakeholder consultation to better understand current capacity among the broad stakeholder base. Experts are being engaged to establish the method for measuring environmental impact to be used in the environmental risk mitigation planning. A workshop to rate the environmental impact of Acacia pests is planned in November 2019 involving staff from NAQS and other subject matter experts.

Collaborator: Plant Health Australia. Investment: \$74,566. Timeframe: Apr 2019 - Jun 2020.

Environmental Biosecurity Plan for Mangroves and Associated Communities

Mangroves are critical parts of coastal ecosystems and deliver a range of ecosystem services, including erosion control, storm protection, waste treatment, carbon sequestration, and are breeding and feeding grounds for fish and other species. Many are close to first points of entry of ships and aeroplanes. This project develops an environmental risk mitigation plan for mangroves and associated communities, including (i) determining the biosecurity risks, risk pathways and review risk mitigation actions with input from external stakeholders; (ii) an environmental biosecurity risk mitigation plan; and (iii) deliver a stakeholder workshop to map the various roles and responsibilities.

Activities to date include stakeholder engagement; identifying activities currently occurring within mangrove ecosystems; and data sourcing to assist in the analysis of risk pathways. Tasks completed comprise a literature review and the development of a list of exotic biosecurity threats to mangroves and mangrove communities. An informal knowledge sharing workshop with up to four different ranger groups is planned for November 2019 at East Trinity Inlet, Queensland.

Collaborator: Plant Health Australia. Investment: \$117,098 Timeframe: Apr 2019 - Jun 2020.

Upgrading the wildlife disease database

The wildlife health database (eWHIS) is one of the key components of Australia's general wildlife health surveillance system. This project ensures that Wildlife Health Australia can continue the provision of quality database and information services. Data collected in the eWHIS database is provided to the department for national and international reporting (e.g. OIE, IUCN), National Emergency Animal Disease Response plans; situation reports during emergency disease events; and serves as an early warning of diseases that may affect production animals, humans and the environment. An upgrade of

the ageing system will deliver significant benefits to the database's capacity for analysis, reporting and administration management allowing for increased automation of the reporting process and user efficiencies.

Collaborator: Wildlife Health Australia. Investment: \$150,000. Timeframe: May 2019 - Dec 2020.

ExtensionAus - Peri-urban general surveillance in NSW

General surveillance has been identified as a cost-effective form of continuous surveillance for early detection of many pests and diseases. Hence, strengthened general surveillance has been identified as a priority in key strategic biosecurity-related documents, including the Intergovernmental Agreement on Biosecurity (IGAB) and the National Plant Biosecurity Surveillance Strategy 2013-2020. This project will contribute to establishing general surveillance in peri-urban areas through the creation of an online Community of Practice to improve small landholder and community interest group involvement in general surveillance in the Greater Sydney and other peri-urban areas in NSW.

To date this project has secured involvement from representatives of various stakeholder organisations and experts to form the Community of Practice. A bootcamp and a planning meeting have been held to outline the scope and focus of the project and identify potential communities of interest, that is, community groups that could be engaged to participate in data collection and reporting activities. The Community of Practice's website is expected to be launched at the end of 2019.

Collaborator: New South Wales Department of Primary Industries. Investment: \$244,348. Timeframe: Jun 2019 - Jul 2022.

Termite delimitation survey of Christmas Island – *Coptotermes gestroi* and *Coptotermes michaelseni*

Coptotermes gestroi and *Coptotermes michaelseni* are termite species that are invasive, destructive, and exotic to Christmas Island. Christmas Island is a Commonwealth External Territory and hence the Commonwealth is responsible for taking the lead on biosecurity incident responses on the island. This project involves a survey to collect adequate data to support the National Biosecurity Management Consultative Committee's decision on whether *C. gestroi* and *C. michaelseni* are technically feasible to eradicate from Christmas Island.

Collaborator: Western Australian Department of Primary industries and Regional Development. Investment: \$45,455. Timeframe: May 2019 - Oct 2019.

Engage Natural Resource Management organisations in environmental biosecurity

The regional Natural Resource Management (NRM) sector has significant capacity and capabilities that could contribute to environmental biosecurity. This project delivered a workshop with senior officials from NRM organisations on 26 June 2019 in Sydney to introduce biosecurity, including what biosecurity is, the challenges and successes. The workshop explored the alignment between

environmental biosecurity and NRM and where the NRM organisations can assist in surveillance and responding to environmental biosecurity threats. It also initiated high level discussion between those already involved in the biosecurity system and the NRM organisations.

Collaborator: Queensland Regional Natural Resource Management. Investment: \$22,727. Timeframe: May 2019 - May 2019.

Supporting the National Invasive Ant Biosecurity Plan implementation

Ants detected in and around Australia's major ports as part of on-going surveillance activities need to be quickly identified to prevent the establishment and spread of invasive ants. A key necessity is to quickly distinguish exotic ants from native and endemic ants commonly found around ports. It is therefore important that ants commonly encountered around ports are easily identifiable. Action 1.2 in the National Invasive Ant Biosecurity Plan 2018-2028 stresses the importance of having resources available for the identification of native and exotic ant species that can be used by front-line biosecurity staff in the Department of Agriculture.

Production of high quality ant images

This project contributed images for five ant species to work already commissioned elsewhere in the department involving the production of a series of high resolution images of ants commonly found around ports.

Collaborator: Queensland Museum. Investment: \$18,182. Timeframe: May 2019 - Jun 2019.

Production of supporting material for high quality ant images

This project contributed supporting information for images of 10 ant species commonly found around major ports. Combined, the images and supporting information will be used to readily exclude ant species commonly encountered around ports when ant detections are made.

Collaborator: Magee Consultancy Services. Investment: \$45,000. Timeframe: May 2019 - Jul 2019.

Providing sponsorship funding for the 2019 Australian Biosecurity Symposium

This sponsorship provided the department with all the inclusions of a Diamond Sponsorship package for the 2019 Australian Biosecurity Symposium that was held 12-13 June. It provided an opportunity to recommend speakers who align with the department's biosecurity priorities, including environmental biosecurity, and create awareness of key biosecurity issues among the wide variety of 300 attendees. The sponsorship contributed to environmental biosecurity education and communication.

Collaborator: Animal Health Australia. Investment: \$32,727. Timeframe: Jun 2019 – Jun 2019.